Ray-Based Semantic Evidence Retargeting for Multi-View 3D Editing

Anonymous manuscript

Abstract

An edited image and undeformed source views describe different geometric states of the same object. Direct multi-view projection can therefore deliver appearance features to an incorrect region after deformation. We study a feature retargeting path that retains the native target conditioning and adapts source evidence through camera-ray groups. A frozen semantic matcher provides correspondence proposals; the geometry stage shares their weighted evidence across each ray and uses a gated source-contrast retargeter. The appearance stage uses a separately evaluated Top-1 retargeter on generated geometry. We distinguish retrieval diagnostics, matched geometric ablations, and fixed-geometry appearance comparisons, preserving the pretrained function at initialization. The evaluation separates source appearance, target alignment, and global image-distribution endpoints, with source-cluster uncertainty and a development-only model freeze. On 508 held-out edits from 134 source clusters, the frozen selected pipeline improves vanilla fine-tuning in source LPIPS (-0.00104), target PSNR ($+0.192$ dB), SSIM ($+0.00116$), and mask IoU ($+0.0120$); all four predeclared 16-endpoint family-adjusted intervals exclude zero. Surface CD and F-Score have favorable point estimates but unresolved intervals, while DreamSim and CLIP do not establish a preservation gain. None of the eight adjusted selected-versus-Top-1-retargeting intervals excludes zero. Thus the held-out evidence supports specific improvements over vanilla fine-tuning, while leaving the extra contribution of the selected matching and aggregation path unresolved.

1 Introduction

Consider editing an object's shape from a single target image while retaining appearance evidence from its original views. The edited target and the sources depict different geometric states. Even with calibrated cameras, a source feature projected along its original ray need not describe the desired edited region. A pretrained multi-view generator supplies a useful 3D prior, but its native spatial associations do not resolve this intentional conflict.

Semantic correspondence can reconnect related regions across the edit. However, selecting one match can discard a useful alternative, while mixing many matches can transfer irrelevant values. A better retrieval score also does not establish that the generated surface or material improves. The delivery path matters: adapted evidence enters an existing projection-based conditioning function whose target contribution and source features already have a trained numerical scale. Figure 1 illustrates the distinct target and source evidence and the evaluated view roles.

For paired supervision, we construct DeformNeXt-10k using donor-guided GeoStyle deformation with source assembly constraints and final-pair screening. The dataset retains exact canonical mesh correspondence while providing 10,165 accepted edits from 3,483 source assets, rather than one fixed edit per object. Section 4.1 describes the construction, split boundary, and limitations of its shortened production optimization budget.

We study source adaptation within that native function. Voxels projecting to the same source-camera patch share one evidence decision. A frozen matcher constructs semantic proposals from the observed target members of that ray; the selected SS design aggregates native values without a new value/output projection. A source-contrast retargeter then supplies a gated correction to the native carrier. Its zero initialization preserves the original function, and rays without proposal support retain the pretrained prior. The target conditioning stream is retained; exact target output preservation remains an evaluation question.

Our experiments distinguish stage objectives. SS candidates are evaluated for geometric alignment under common frozen downstream models. PBR candidates are evaluated for source appearance on identical generated geometry, guides, noise, and rendering triangles. This produces a measured composition of Fixed Top-K SS and a separate Top-1 retargeting PBR stage, rather than requiring one aggregation operator to be best at both tasks.

The contributions are an explicit ray-shared source-adaptation formulation; matched tests separating raw retrieval, decoder descriptors, scalar scores, aggregation, and retargeting; and a stage-specific evaluation with a frozen held-out protocol. Candidate coverage, component effects, and final pipeline quality are separate claims. We retain negative results and uncertainty instead of inferring necessity from the architecture alone.

2 Related work

Generation and editing. Sparse representations support high-resolution geometry and materials in TRELLIS.2 [20]; Pixal3D introduces pixel-aligned projection conditioning [10]. Reconstruction–generation combinations include ReconViaGen [2], RecGen3D [9], and Gen-

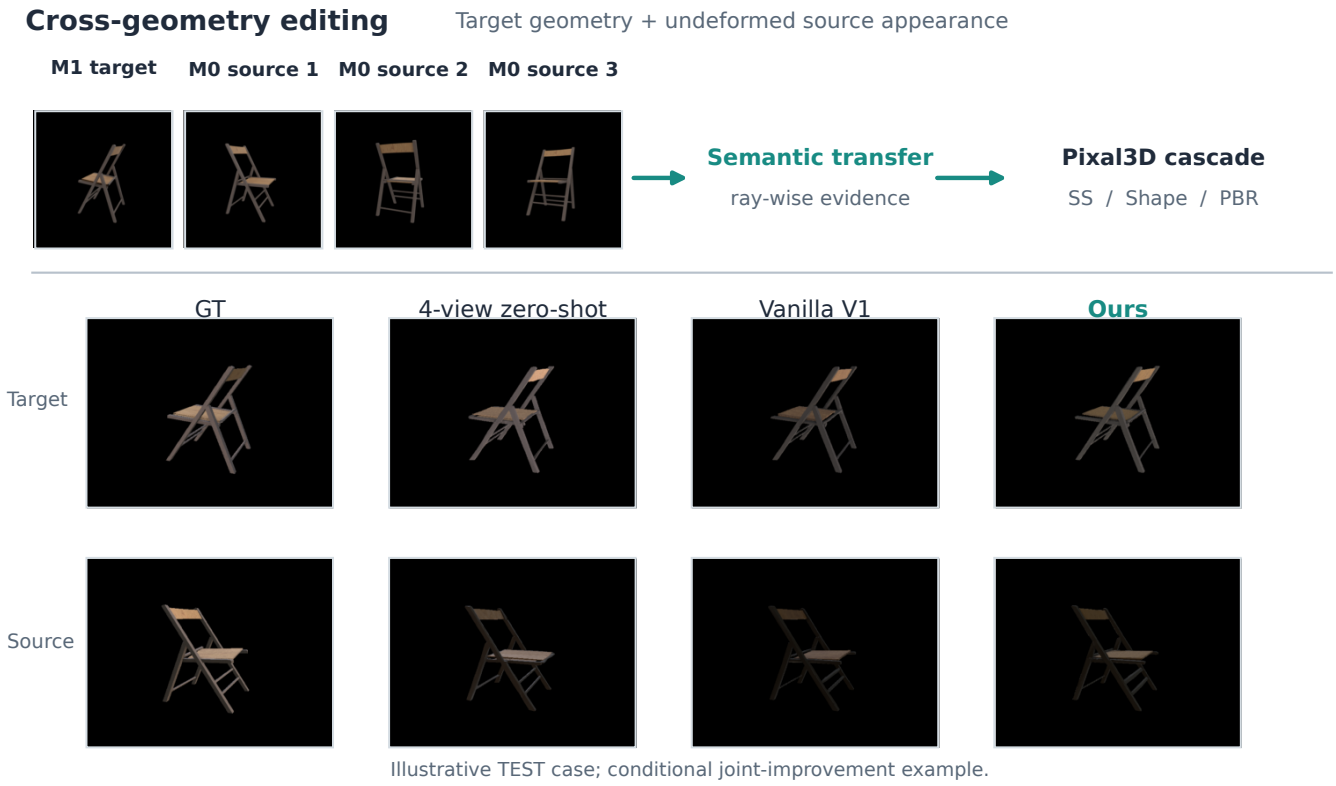


Figure 1: Cross-geometry editing from one edited target RGB view and three undeformed source RGB views. The lower rows compare scored TEST renders at the target camera and one source camera with the common original-material M1 reference. This is the predeclared conditional joint-improvement example, shown to explain the task rather than estimate a population success rate. All compared renders use the archived full-frame pixels and identical cameras.

Recon [13]. We continue the supplied multi-view LoRA implementation rather than claim projection or multi-view conditioning as a new mechanism. Its TRELLIS.2/DINOv3 implementation differs from the published Pixal3D backbone; checkpoint lineage is specified below.

ShapeUP learns source-shape/edited-image translation and uses source renders for texture [6]. GeoStyle deforms a source mesh from an image style reference [3]; APAP [22] and SpaceControl [4] combine geometric control with generative priors. TextureDreamer transfers appearance to a supplied target shape [21]. Our evaluated input is instead edited target RGB plus undeformed source RGB with calibrated cameras, and the output geometry is predicted. Source-view appearance conditioning itself is not our novelty claim, and these input differences preclude an unexecuted quantitative comparison with source-mesh or target-shape methods.

Correspondence and aggregation. MV-Adapter supports multi-view generation [8]. MVDiffusion exchanges correspondence-conditioned information and uses zero-initialized residual layers [15]; this initialization principle is a safeguard we reuse. CorrAdapter aggregates matched regions for multi-image consistency [25]. Our question concerns semantic evidence delivered into a 3D projection-conditioning path when the observed images intentionally depict different geometries.

Diffusion and DINO features expose complementary correspondence properties [14, 23]. VGGT-Ω learns geometry-aware representations [17], and RoMa-Ω separates feature similarity, learned probes, and dense matching [11]. Its co-visibility confidence is distinct from our categorical weighting over retained proposals. These works motivate separating descriptor quality, matching, aggregation, and generative effects; they do not establish those effects in our editing task.

3 Method

3.1 Native carrier and the sharing unit

Let T be the target feature map, S_j a source map, and P_j its calibrated bilinear projection. We retain the native four-view carrier C_{MV4}, including its target contribution and global tokens. For source camera j and image patch p, the sharing unit is

$$\mathcal{R}_{j,p} = \{x : \pi_j(x) = p\}. \tag{1}$$

Grid sites on this camera ray share an image-space evidence decision. The decision is not an independently

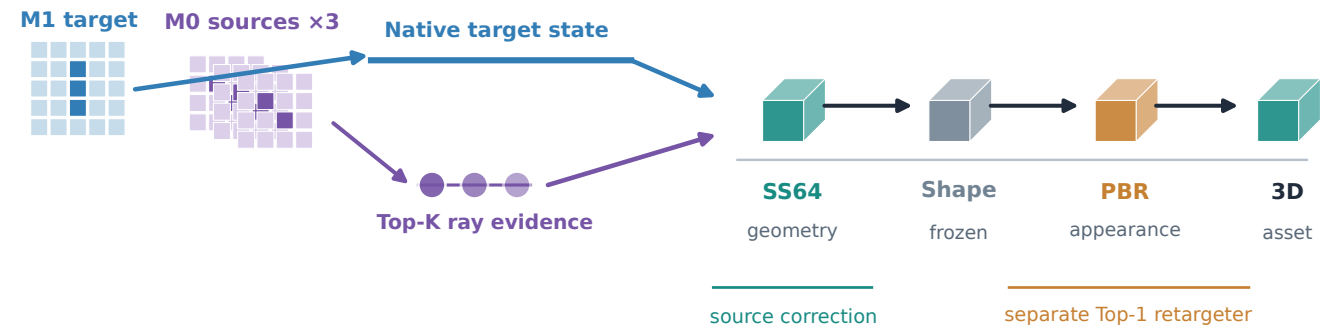


Figure 2: Stage-level overview. The edited target follows the native Pixal3D conditioning path. Semantic evidence from undeformed source views corrects the SS geometry carrier; Shape512/1024 are frozen. PBR uses a separately selected Top-1 source retargeter for appearance. Figures 3 and 4 explain the correspondence and source correction.

learned attention output for each voxel feature. The SS implementation uses a 16^3 conditioning grid and a 32×32 patch map; SS64 denotes decoded occupancy resolution. Camera projection defines membership without target surface depth or a surface-visibility write mask at inference.

3.2 Semantic proposals and matching scores

Each observed target patch m retrieves 64 semantic proposals across the three source images. A frozen D1 decoder contextualizes its descriptor and produces scores s_{mk}. The decoder has one query-only cross-attention block and a bounded scalar residual head. Matching keys are 1024-dimensional layer-12 FRAME Omega descriptors; SS values are separate, 1024-dimensional native DINOv3 features.

Topology-derived barycentric correspondence and training-only depth checks provide valid positive patch IDs $\mathcal{P}_m$. Given raw proposals $\mathcal{C}_m$, the decoder minimizes

$$\ell_m = -\log \frac{\sum_{k \in \mathcal{C}_m \cap \mathcal{P}_m} e^{s_{mk}}}{\sum_{k \in \mathcal{C}_m} e^{s_{mk}}}. \tag{2}$$

Only queries with a positive proposal contribute. Proposal misses and unobservable queries are recorded separately. Decoder training uses 9,143 certified training examples and seven passes; Omega and the complete matcher are frozen during subsequent generative fine-tuning. Topology/depth labels are supervision and scoring data, not inference conditioning.

Distinct observed target patches in a source ray form its member set $\mathcal{M}_r$. Their normalized distributions vote once per member:

$$\bar{p}_r(k) = |\mathcal{M}_r|^{-1} \sum_{m \in \mathcal{M}_r} p_m(k), \tag{3}$$

$$\mathcal{K}_r = \mathrm{Top64}(\bar{p}_r), \qquad p_r(k) = \frac{\bar{p}_r(k)}{\sum_{l \in \mathcal{K}_r} \bar{p}_r(l)}. \tag{4}$$

Missing member entries have zero mass. Scores use softmax temperature 0.05. This retains complementary modes before truncation instead of matching one averaged descriptor. Input alpha and camera addresses define observation support; they are not surface co-visibility. Empty members or insufficient valid proposals abstain. A concentrated p_r need not be correct: native fallback detects missing support, not every confident mismatch.

3.3 Matching-aware aggregation and retargeting

The selected SS stage aggregates untransformed native values,

$$E_r = \sum_{k \in \mathcal{K}_r} p_r(k) v_k. \tag{5}$$

Carrier correction, source use, and retargeting consume this same evidence. No additional learned Q/K or V/output head is required by this instantiated design. Such heads are evaluated as alternatives, not declared necessary.

With learned source-use probability u_r and signed bounded gate g,

$$R_r = R_\phi([E_r - S_r, T_r]), \tag{6}$$

$$\Delta_r = u_r\{g(E_r - S_r) + R_r\}, \tag{7}$$

$$C = C_{\mathrm{MV4}} + \tfrac{1}{4} \sum_{j=1}^{3} P_j(\Delta_j). \tag{8}$$

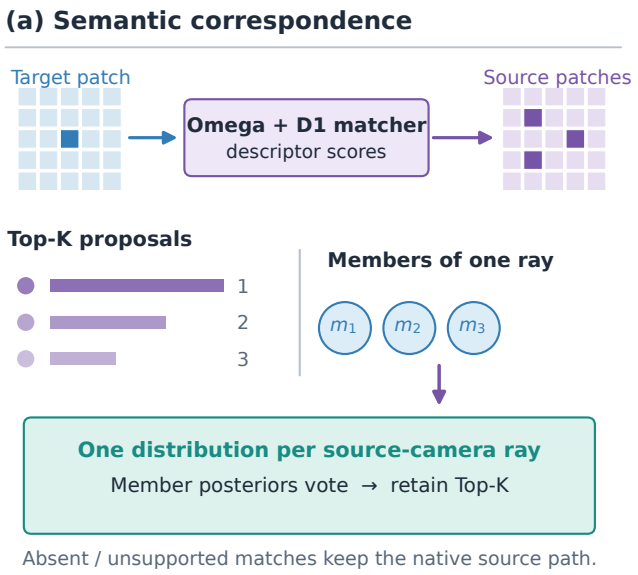


Figure 3: Frozen semantic correspondence. Omega descriptors generate source proposals and the D1 decoder scores them. Distinct observed target patches sharing a source-camera ray vote into one Top-K posterior. Matching keys are separate from the native source values used for generation.

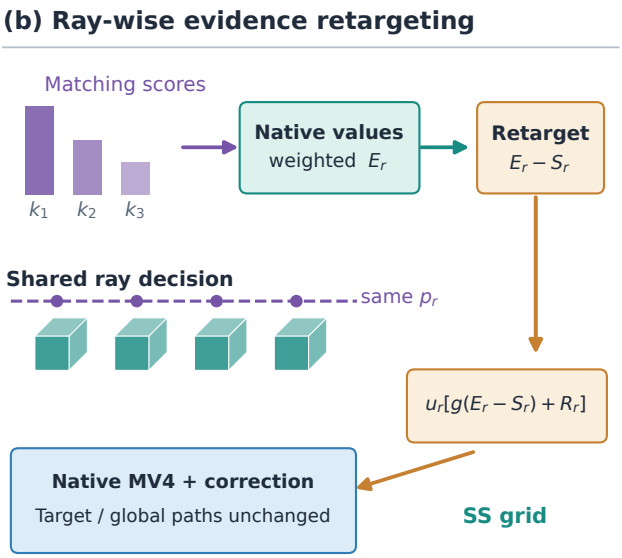


Figure 4: One source-camera ray shares its posterior across intersected grid sites. The selected SS path weights native source values, forms a bounded source-contrast correction, and adds that correction to the native MV4 carrier. The target contribution and global tokens remain in the carrier; unsupported rays retain the native source path.

T_r pools native target context over the ray's grid sites; the distinct observed members construct the matching distribution. Subtracting S_r accounts for its existing contribution to the carrier. Initially $g = 0$ and the retargeter output is zero, recovering native conditioning. Unsupported rays retain native source features rather than receive invented zero-valued observations. Target feature extraction and its direct conditioning path are retained, but exact generated target RGB preservation is not guaranteed.

3.4 Stage-specific composition

SS64 and PBR1024 continue their supplied multi-view LoRA checkpoints; Shape512/1024 stay frozen. SS uses the weighted Fixed evidence above. The selected appearance stage uses its separate Top-1 retargeter on generated geometry. Its values are native DINOv3/NAF features. Neither the PBR feature variants nor the SS learned-refinement variants are silently substituted for this selected composition. Stage choices are based on their respective geometric and fixed-geometry appearance development comparisons.

4 Experimental setup

4.1 DeformNeXt-10k construction

We construct paired shape edits from the textured assets and hierarchical part annotations of PartNeXt [18]. A canonical source $M_0 = (V_0, F, \mathcal{A}_0)$ defines vertex indices, ordered faces, and the UV/material/corner-attribute authority. Each accepted variant is $M_{1,k} = (V_{1,k}, F, \mathcal{A}_0)$: positions change while correspondence and face connectivity/order remain fixed. Shared face indices and barycentric coordinates therefore identify corresponding source and edited surface points without nearest-neighbor reassignment.

Donor-guided proposals. We select donors from the same category using frozen semantic-role and structural-compatibility heuristics. Semantic roles can contain multiple disconnected pieces; matching individual leaf meshes or requiring equal part counts would unnecessarily restrict shape diversity. Source/donor identities and configuration hashes are recorded before each run. Eight original-appearance donor renders personalize an SDXL prior with GeoStyle [3]. The donor is a morphology prior, rather than target geometry or an input to the retargeting model.

GeoStyle proposes face-Jacobian changes under image-space score distillation. Our production path adds a category prompt and source assembly constraints to the reconstruction: selected primary surfaces deform, complementary physical components follow rigid motions, and source-derived attachment proxies couple their movement. These contacts are geometric estimates, not annotated physical joints. The main production configuration uses 800 donor-specific LoRA updates and 300 deforma-

Table 1: Frozen DeformNeXt splits. Variants of one source and geometry cluster stay together; category proportions are not forced to be uniform.

Split	Edits	UIDs	Clusters	Classes
Train	9,149	3,184	1,790	50
Validation	508	153	132	42
Test	508	146	134	42
Total	10,165	3,483	2,056	50

tion iterations; donor LoRA and source preprocessing are reused across pairs. This shortened configuration was adopted for throughput and is not demonstrated equivalent to the official 3,600-iteration configuration. Supplementary details give the other changes and retained early configurations.

Final-pair selection. The category-prompt/assembly proposal (PA) and an optional 120-iteration Jacobian, crease-aware normal, and neighboring-displacement refinement (balanced JN-L) are separate candidates. Each is assessed anew against M_0. Agreement with the initial proposal or the donor is not an admission criterion. We check canonical correspondence, finite geometry and catastrophic failures, meaningful deformation, paired RGB usability, native-representation survival, and actual training-loader compatibility. Multi-view model-assisted screening identifies obvious spikes, detached components and assembly collapse; it is not independent human validation. Small intersections, gaps and folds are recorded as diagnostics rather than universally rejected. The dataset is therefore not certified intersection-free. Both meshes retain one source-defined 0.48 frame and are rendered with the source materials and matched cameras; edited meshes are not independently normalized. All approved distinct variants are retained, with near-identical final geometries merged.

Scope and splits. The final snapshot contains 10,165 distinct deformations from 3,483 source UIDs across 50 categories, averaging 2.92 variants per source. This is a deformation count, not 10,000 independent source objects or an estimate of yield over all PartNeXt assets. After construction, a fixed seed assigns approximately 90/5/5 of the examples to train/validation/test while keeping every source UID and source geometry cluster in one split (Table 1). Donor disjointness is not enforced; the held-out evaluation measures retargeting on admitted edits from this construction process, not generalization to an unseen deformation generator.

4.2 Training and evaluation protocol

Data and initialization. We use the frozen splits in Table 1. PBR training excludes ten recorded oversized training examples, using 9,139; no test asset is excluded by that training limit. Source UID and source-asset clusters are disjoint across these deformation splits. Overlap with earlier multi-view pretraining is not established. The earlier mixed385 subset of 80,892 training and 950 validation assets is pretraining lineage, not the deformation fine-tuning set.

We load the original configurations and complete base+LoRA weights: SS64/Shape512 step 4,210 and Shape1024/PBR1024 step 16,850. The matching backbone is VGGT-Omega epoch 7. SS uses 480 updates, global batch 96, LoRA rank/alpha 16, and denoiser/conditioner learning rates $10^{-5}/10^{-4}$. PBR uses 1,900 updates, global batch 24, and rates $5 \times 10^{-6}/10^{-4}$. These are approximately five training passes. Ablations retain paired inputs, initialization, budget, sampler, and noise. Removed modules/channels are physically absent, not filled with zeros.

Development and freeze. SS diagnostics use DEV128/93 sources and three paired generation noises, averaged within an example. Surface ablations use the same 128 examples, one noise, frozen Shape and common Top-1 R PBR. PBR appearance uses DEV32 with identical generated geometry, archived guides, texture noise, actual rendering triangles, and references. SS selection uses CD/F-Score; PBR selection uses source LPIPS/DreamSim with complementary CLIP. Cross-SS appearance is a secondary diagnostic, not a veto of geometric improvement.

Two independent SS training seeds for V1/R/Fixed and two PBR seeds for vanilla/R are reported separately. The SS confirmation concerns occupancy; PBR confirmation is a four-example GT-shape target-camera training diagnostic. Neither is two-seed full-pipeline quality confirmation. Two fresh loads on VAL36 verify the selected SS checkpoint, archived latent/occupancy arrays, prefix32, and the final batch of four. Actual weight/config/cache bindings and the complete evaluation protocol are frozen before TEST inference.

Comparators and generation. The frozen inferential TEST comparison contains five physical pipelines: original source-only RGB4; original edited-target RGB1+source RGB3; vanilla V1; Top-1 R; and selected Fixed SS+Top-1 R PBR. Source-only includes the original M0 view at target camera8, not just three sources. Edited inputs are M1 RGB at8 plus M0 RGB at9/10/11. Shape weights are common. Original zero controls retain original PBR; V1 uses vanilla 1900 PBR and R/selected use Top-1 R 1900. SS generation uses batch 8, 12 steps, guidance 3 and the prescribed per-asset noise map. Native downstream sampling configurations are retained.

Table 2 also includes a post-hoc descriptive input reference: the original Pixal3D single-view SS, Shape512, Shape1024, and PBR denoiser checkpoints, with the project's calibrated projective DINO conditioner, receive

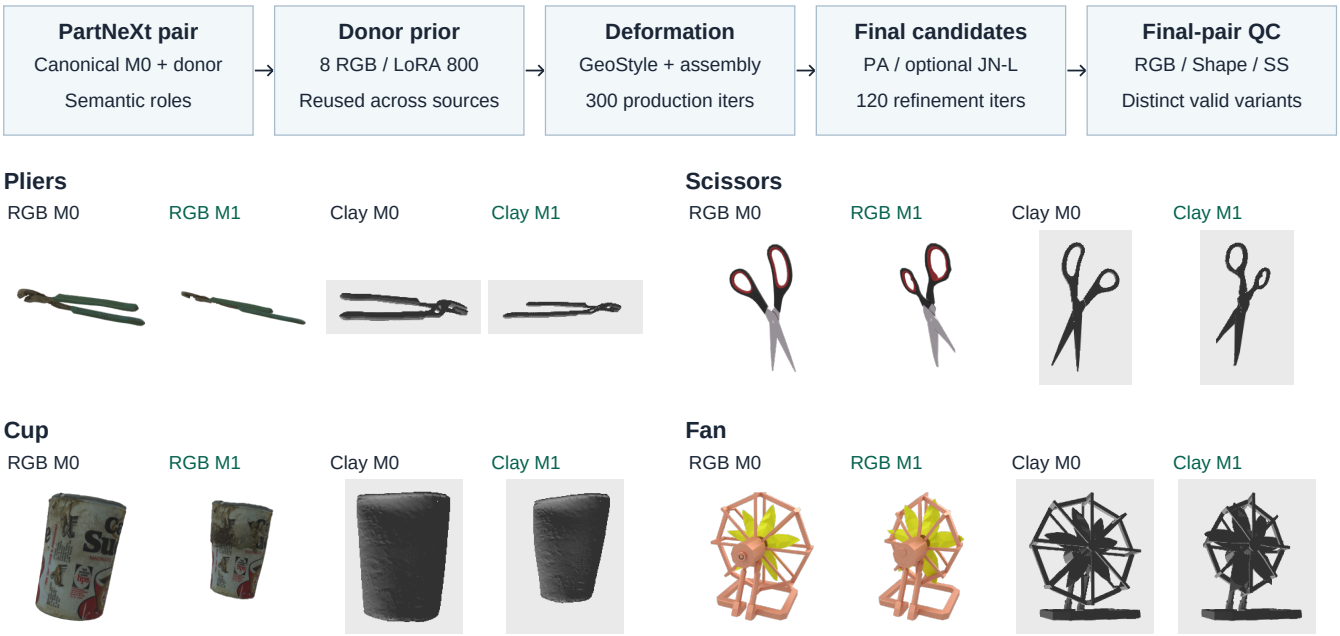


Figure 5: DeformNeXt construction and approved source/edited pairs. Donor images guide morphology; source assembly constrains reconstruction, and each final candidate is screened independently. Examples show archived RGB (camera 8) and clay (camera 14), with a common foreground-union display crop and scale for M0/M1 at each camera. Four categories are drawn from a pre-existing large-edit illustration selection whose membership was checked against the final snapshot. They illustrate visible changes, not typical quality, generator yield or a 300-versus-3,600-step comparison.

only the edited M1 target RGB at camera 8. This is a project-specific 1024-cascade evaluation, not the official Pixal3D inference stack. It uses the same 508 TEST examples, per-asset seeds, scoring cameras, native renderer, and original-material M1 references as the five-model comparison. The pretrained single-view weights retain their 12-step sampler defaults, including SS guidance 7.5. There is no multi-view or deformation fine-tuning. It was evaluated after the five-model freeze, so it is not included in the adjusted comparison family.

Metrics and references. Source LPIPS [24], DreamSim [5], and separate OpenAI CLIP image cosine [12] average cameras 9/10. Target PSNR/SSIM [19]/2D mask IoU use 8. References use original materials on edited M1 geometry, not undeformed source pixels. Surface CD is the symmetric mean unsquared Euclidean nearest-point distance, using 20,000 area-weighted samples. F-Score uses threshold 0.02 in the common source-defined normalized frame [16]; there is no per-output ICP alignment. Voxel IoU is diagnostic. Full-frame image scores are primary; foreground/GT-only crops are supplementary.

FID [7]/KID [1] use the same 1,524 images per model/reference, cameras 8–10, official Inception 2048 FP32 features, and torch-fidelity 0.3.0 float64 reductions. KID uses 100 subsets of 1,000 and seed 42017, retaining signed raw values. Distribution differences are descriptive; subset variability is not a source-cluster interval. These are camera-8/9/10 evaluations, not evidence of general novel-view performance. The RGB1 reference sees only camera 8, so cameras 9/10 are held out for that row; the four-view pipelines receive all three scoring cameras as inputs. This table uses the project's native mesh renderer and black-background image compositing. The separate external-baseline benchmark uses Blender/Cycles, scores image metrics on cameras 9/10, and uses a different FID/KID pool and surface-sampling protocol. Its scores are not directly comparable here.

GT raster quality control uses one-pixel edge tolerance and a uniform whole-pixel rounding of its 1% allowance. The post-TEST correction and GT-only audit are disclosed in the Supplement; metric formulas are unchanged.

Statistical scope. Means weight deformation examples equally. Paired bootstrap resamples source clusters, retaining within-source dependence; equal-source sensitivity is also retained. The development downstream family is six contrasts across eight endpoints, with 48,000 draws and 48-comparison Bonferroni intervals. Its geometry endpoints guide SS interpretation; this family is not reduced after observing results. Stage IoU has its separate six-contrast family. The selected TEST model versus V1/R uses 4,000 draws with seed 20261022, eight pairwise endpoints and a 16-comparison family across 134 sources. Intervals condition on frozen checkpoints and generation, not a population of training seeds. A zero-containing in-

terval does not prove equivalence.

5 Results

5.1 Frozen held-out comparison

Table 2 reports the complete frozen five-pipeline comparison plus the added single-view reference on 508 TEST objects and 134 source clusters. The added RGB1 row has the highest target input-camera PSNR/SSIM/mask IoU points (24.543 dB/0.95139/0.92725), but its surface CD and F-Score (0.01789/0.74311) are worse than the four-view edited zero-shot and selected models. Its source LPIPS/DreamSim/CLIP are 0.06090/0.16202/0.94062, versus 0.05620/0.15207/0.94302 for the selected model. The calibrated single-view pipeline differs in denoisers and guidance from the multi-view controls, so these descriptive differences cannot be attributed to view count alone. It also differs from the separately evaluated official Pixal3D baseline in camera estimation and cascade resolution; their numerical results belong to different protocols. Within the frozen five-model family, supplying the edited target RGB changes the zero-shot target PSNR from 22.406 to 23.542 dB, mask IoU from 0.8149 to 0.8825, CD from 0.01674 to 0.01488, and surface F-Score from 0.7662 to 0.8083. Vanilla fine-tuning reaches 23.836 dB, 0.8890, 0.01411, and 0.8285, respectively. These zero-shot and vanilla comparisons are descriptive; the predeclared adjusted family tests the selected pipeline against the two fine-tuned controls. The different zero-shot inputs separate access to edited RGB evidence from the subsequent fine-tuning comparison, without treating input availability as a learned module effect.

The selected Fixed-SS/Top-1-R-PBR composition has source LPIPS 0.05620 and target PSNR 24.028 dB, SSIM 0.94753, and mask IoU 0.90107. Relative to vanilla V1, the paired differences are −0.001044, +0.1923 dB, +0.001156, and +0.012041. Their 16-endpoint family-adjusted 95% intervals exclude zero (Supplementary Table 18). The selected pipeline has a lower CD point estimate (0.01360 versus 0.01411) and a higher surface F-Score (0.83453 versus 0.82847), but the adjusted intervals for both contain zero. Source DreamSim is 0.15207 versus V1's 0.15146, and its adjusted interval also contains zero; the same is true for source CLIP. Thus the held-out support is endpoint-specific, rather than a claim that geometry and source appearance improve on every measure.

Against the Top-1 retargeting baseline, the selected pipeline has small favorable points for LPIPS, target PSNR, SSIM, mask IoU, and CD, but a higher (worse) DreamSim and slightly lower F-Score. Every one of the eight predeclared adjusted intervals contains zero. This comparison does not establish that the selected Top-K weighted SS path contributes additional held-out quality beyond the retargeting baseline, nor does it establish equivalence. The added single-view baseline has the lowest FID point (34.147), while the source-only four-view zero-shot model has the lowest KID point (0.001491). FID and signed raw KID summarize common 1,524-image camera-8/9/10 pools and are descriptive, without a paired significance claim.

5.2 Geometric component comparisons

All seven SS designs have completed 480 updates, DEV128 three-noise diagnostics, and actual surface evaluation under common downstream models. Spatial, raw semantic, decoder-cosine and scalar Fixed give CD 0.0129704, 0.0128789,0.0128065 and 0.0127572, with F-Score 0.839333,0.841031,0.842414 and 0.844058. The point-estimate ladder improves both geometric endpoints, but every interval in the 48-comparison family includes zero.

Fixed minus Top-1 changes F-Score by 0.003963, with adjusted interval $[-0.000910, 0.009748]$; Fixed minus Uniform gives 0.001553 with $[-0.003326, 0.007439]$; physical retargeter removal gives a Fixed advantage of 0.009020 with $[-0.001582, 0.020293]$. CD intervals also contain zero. We select Fixed by the lowest CD/highest F-Score points, while retaining uncertainty about component effects. Spatial 3 versus semantic 64 is a module substitution, Top-1 versus weighted Top-K bundles candidate count and aggregation, and scalar matching can change support. Uniform on the same support is the separate weighting control.

5.3 Appearance on identical selected geometry

The four PBR variants share actual generated Fixed geometry, native guides, texture noise, triangles and references. Top-1 R has the lowest source LPIPS/DreamSim points, 0.053290/0.157828, versus vanilla 0.053513/0.158346. Their exploratory paired deltas are −0.000223 with $[-0.000539, 0.000035]$ and −0.000517 with $[-0.002250, 0.001183]$. Both intervals include zero. Feature/Raw variants do not improve either primary point over Top-1 R; vanilla has the best complementary CLIP point. We choose the measured Top-1 R PBR and report target image metrics separately. Identical CD/F-Score confirm geometry equality, not a PBR geometry gain. The feature/Raw arms retain a Top-1 carrier/use path and do not establish the effect of a shared Fixed PBR pathway.

5.4 Retrieval evidence and generated quality

Decoder matching improves exact rank-one coverage from 1.25% to 7.73% in the separate DEV32 diagnostic. Member proposals raise exact@64 coverage from 50.76% to 73.36%. Native evidence-use analysis finds 31,408 observable Top-1-miss/Top-K-hit relations; Fixed places

Model	Source preservation			Global naturalness		Target alignment				
	LPIPS↓	DreamSim↓	CLIP↑	FID↓	KID↓	PSNR↑	SSIM↑	M-IoU↑	CD↓	F-Score↑
Pixal3D RGB1 (calibrated)	0.060898	0.162024	0.940617	34.147430	0.0016813205	24.5428	0.951388	0.927251	0.0178908	0.743113
Zero-shot, source-only RGB4	0.061884	0.153975	0.943278	36.089871	0.0014905934	22.4062	0.936489	0.814896	0.0167443	0.766221
Zero-shot, edited target + source RGB3	0.058324	0.154276	0.942090	35.024202	0.001618189	23.5416	0.945824	0.882545	0.0148793	0.808255
Vanilla V1	0.057245	0.151460	0.942854	34.658819	0.0017537581	23.8360	0.946375	0.889032	0.0141140	0.828473
Top-1 R	0.056329	0.151162	0.943177	34.930351	0.001822029	23.9966	0.947295	0.898210	0.0136234	0.834758
Selected model	0.056201	0.152069	0.943018	34.361368	0.0016011107	24.0283	0.947532	0.901072	0.0136001	0.834532

Table 2: Internal TEST508 comparison (508 edits; 134 source clusters). The first row uses original Pixal3D single-view denoisers in our calibrated projective 1024-cascade pipeline, with edited target RGB only; it is a post-hoc descriptive reference, not the official Pixal3D baseline. Source metrics use cameras 9–10, which are held out for RGB1 but inputs to the four-view rows; target metrics use input camera 8. CD and F-Score use a common normalized frame (F@0.02). FID/KID use identical 1,524-image original-material M1 reference pools; KID is signed raw units. External baselines scored with Blender/Cycles under a different protocol cannot be inserted directly.

Working Fixed-context SS design	CD↓	F-Score↑	SS IoU↑
A. Matching ladder: compare adjacent rows			
Original spatial3, RAW cosine	0.0129704	0.839333	0.472267
Raw semantic64, cosine	0.0128789	0.841031	0.472975
Decoder semantic64, cosine	0.0128065	0.842414	0.473450
Decoder semantic64, scalar (Fixed)	0.0127572	0.844058	0.473471
B. Aggregation and retargeting: compare with Fixed above			
Actual Top-1 feature (No-score)	0.0127795	0.840095	0.472600
Uniform on the same64 support	0.0128304	0.842505	0.473613
Physical retargeter removal	0.0129562	0.835038	0.468832

Table 3: Completed matched development ablation. All seven rows have final SS and cascade results, with the same actual SS checkpoint binding in both evaluations. SS selection targets geometry: CD/F-Score are primary and IoU is diagnostic. Cross-SS appearance scores are secondary and do not veto geometry gains. SS IoU uses DEV128 and three paired noises; geometry columns use the same128 assets and one noise, with frozen Shape and common Top-1 R PBR. Panel A separates raw semantic retrieval, decoder descriptors and scalar scores in the same Fixed delivery. Spatial3 versus semantic64 is a module substitution, not fixed-K addresses; the spatial row is compared with raw semantic, not treated as a one-change removal from scalar Fixed. Panel B retains the same matching input. Top-1 delivers one actual winner; Uniform changes weighting on unchanged support. Scalar matching can also change support in Panel A. Learned refinement and original ray/member results are reported separately; absent Q/K or deformation-query modules do not receive fictitious ablations.

PBR arm on identical Fixed geometry	LPIPS↓	DreamSim↓	CLIP↑	Target PSNR↑	Target SSIM↑
Vanilla	0.053513	0.158346	0.942179	24.47775	0.951534
Actual Top-1 + R	0.053290	0.157828	0.941692	24.54433	0.951563
Top-1 + learned feature + R	0.053326	0.157985	0.941759	24.51012	0.951536
Raw Top-K + learned feature + R	0.053361	0.158261	0.941559	24.52268	0.951539

Table 4: Completed appearance-stage comparison on32 development examples using the selected Fixed SS geometry. Only PBR changes: the generated mesh, native guides, initial texture noise, sampler, rendering triangles, and material references are identical. Top-1 + R has the lowest source LPIPS/DreamSim points; we select its actual1900-step checkpoint. Its vanilla-comparison intervals for both primary distances include zero. Vanilla has the highest complementary CLIP point. Target image metrics are reported separately; CD0.0123276 and F-Score0.856566 are identical for all four rows and establish no PBR geometry gain. Feature/Raw retain the Top-1 carrier/use path and do not test a shared Fixed PBR pathway. These development results are not held-out confirmation.

more probability on GT proposals than post-hoc Uniform. These measurements establish available evidence and its allocation, not a generated-quality improvement. The original learned ray-versus-voxel and member-versus-mean studies do not establish positive quality effects; their scope differs from the selected identity-value path. Full diagnostics and the actual annotated example are in the Supplement.

5.5 Prespecified qualitative comparison

6 Discussion and limitations

The frozen held-out comparison supports a narrow conclusion: composing the selected geometry and appearance stages improves source LPIPS and three target-image endpoints over vanilla V1 under the declared adjusted

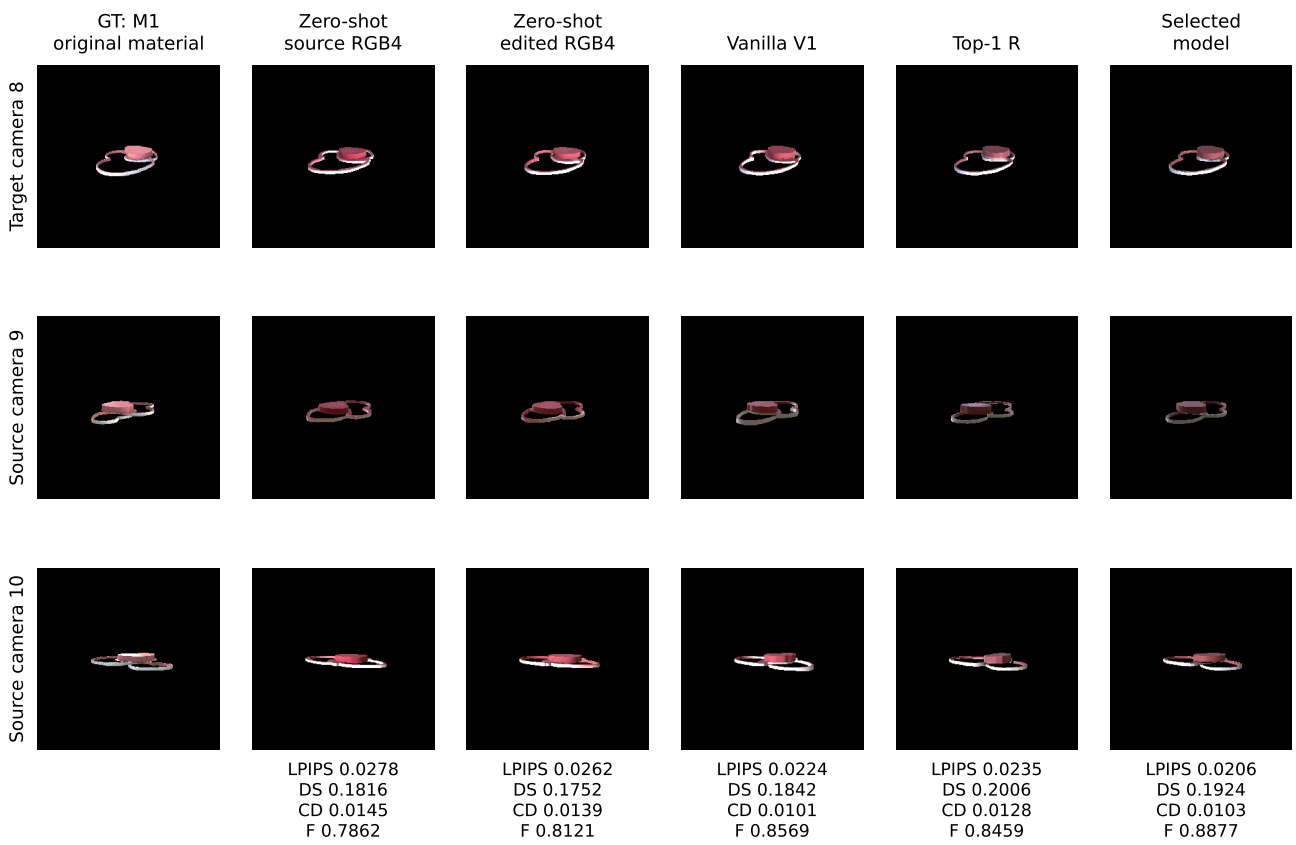


Figure 6: Actual scored frozen TEST example. Rows are target8/source9/source10; columns are the common original-material M1 reference and every frozen model. Selection: outcome-blind. Full-frame panels use the scored archive and official uint8 conversion, without model-specific crop, alignment or rerender. Source annotations use cameras9/10. Conditional categories illustrate their stated conditions and do not estimate population success rates.

family. Surface CD and F-Score remain uncertain despite favorable points. The selected-versus-Top-1-R comparison is unresolved on every adjusted endpoint. These results cannot identify semantic matching, matching score, ray aggregation, or the retargeter as individually necessary. The completed DEV matched studies also leave their component-level 48-comparison intervals crossing zero; retrieval coverage and attention allocation are mechanism evidence rather than generated-quality guarantees.

The outcome-blind panels retain their predetermined samples, while the conditional Supplement includes joint improvement and separate geometry- and appearance-loss cases. Those categories expose failures and are not population success rates. All displayed panels use the scored full-frame archives and common M1 original-material reference. The native target conditioning stream remains intact, but the complete generator does not guarantee pixel-exact target RGB preservation. The PBR selection is a same-geometry appearance comparison; the evaluated PBR feature/Raw alternatives use a Top-1 carrier and do not test a shared weighted Top-K appearance path.

The estimates condition on one frozen full-pipeline checkpoint and one prescribed generation noise per TEST asset. Two SS training seeds and the small PBR training diagnostic do not measure full-cascade training-seed uncertainty. The TEST sources are disjoint within DeformNeXt, but possible overlap with earlier multi-view pretraining has not been audited. Metrics use the specified input cameras; they do not establish novel-view editing quality. FID and KID are renderer-dependent distribution summaries, and their point differences have no source-cluster interval. Foreground and GT-crop sensitivity is reported separately without redefining the full-frame primary endpoints.

Stage objectives and the delivery path. An SS geometric change can alter rendered source appearance even

with a common PBR, so cross-SS LPIPS/DreamSim cannot establish an appearance-module effect. Conversely, fixed-geometry PBR comparisons do not test geometric improvement. We report both groups for the composed system while selecting each stage for its own objective. Native target conditioning is retained, but the output depends jointly on all stages and need not preserve target RGB exactly.

Candidate uncertainty. Higher candidate coverage and GT probability mass need not improve a generated asset. A well-supported ray can still match incorrectly; support is not physical visibility and relative posterior mass is not calibrated correctness. The native fallback handles absent evidence, not every confident mismatch. Additional learned refinement has not established a surface-quality advantage in the completed development comparison. The existence of ray sharing, semantic matching, or attention does not prove their necessity.

Scope of the evidence. The original FULL, its ray/member controls, and the selected shared-evidence design differ in multiple components. Their differences do not isolate shared routing or the final ray unit. Two SS training seeds confirm occupancy trends of combined branches; the two-seed PBR diagnostic uses four GT-shape target examples. Neither quantifies full-pipeline training-seed variability. Test sources are disjoint within DeformNeXt, while overlap with earlier multi-view pretraining is not audited. Input-camera evaluation and renderer-specific FID/KID do not establish novel-view quality or human naturalness judgments. Full-frame appearance scores can be influenced by background; the existing foreground/GT-crop sensitivity is reported separately without changing the primary endpoints.

7 Conclusion

We evaluated semantic evidence retargeting within a frozen multi-view 3D editing pipeline, selecting SS for geometry and PBR for appearance before the held-out test. On TEST508, the selected composition improves vanilla fine-tuning on source LPIPS and target PSNR, SSIM, and mask IoU under the predeclared adjusted comparison. Surface geometry gains and an additional benefit over the Top-1 retargeting baseline remain uncertain. The resulting claim is a measured pipeline-level improvement on specified endpoints, while the separate necessity of matching, weighting, and ray-based aggregation remains an open empirical question.

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A Supplementary evidence and reproducibility

This Supplement retains complete development studies and their limits. It does not reinterpret exploratory results as held-out confirmation.

A.1 DeformNeXt construction details and limitations

Source semantics and assembly. PartNeXt supplies hierarchical semantic labels, not ground-truth mechanical attachments. We map its annotations to the canonical face array and group semantic roles before selecting a deformable primary region. Several legs can share a semantic role while remaining separate physical followers. For reconstruction, exact coincident vertex classes and connectivity define numerical physical units. Nearby source surfaces provide frozen vertex-to-triangle attachment patches, including barycentric coordinates and rest-frame offsets. The default proximity radius is 0.006 times the source diagonal, with at most 96 contact samples per component.

The differentiable reconstruction alternates four projected Poisson/rigid steps; proper rigid rotations use 32 quaternion iterations. A final Dirichlet solve eliminates rigid follower coordinates. Components in assembly islands without a deformable primary are fixed. This is a finite approximation, not an exact nonlinear optimum or a collision guarantee. Internal vertex coalescing is a numerical operation: the exported candidate is restored to canonical vertex indices and checked against the unchanged face array. Canonical source UVs, material assignments and corner attributes remain the rendering authority.

Personalization and optimization budget. The principal donor-reference arm uses eight Cycles RGB images with the donor's original appearance. A clay-reference arm was exploratory, not the default for the full snapshot. The donor-specific DreamBooth LoRA uses 800 updates, rank 16, 512-pixel images, batch one, learning rate 10^{-4}, constant scheduling, no warmup, gradient accumulation one, and text-encoder training. Its caption is *A photo of TOK sculpture*; the production deformation prompt is *A TOK style sculpture of a <category>*. Personalization starts from pretrained SDXL, not from a diffusion model trained from scratch on the construction dataset. Donor LoRA weights can be reused for different sources; source approximate-encoder preparation is also cached independently of pair optimization.

The main production adaptation uses 300 deformation iterations, resolution 512, batch one, and *cage_loss_epoch=83*, with seed 37. The official configuration used as our baseline has 3,600 iterations, resolution 1,024, batch four and a cage-loss interval of 1,000. Main/auxiliary SDS weights change from 10/1 to 2.5/0.25 to compensate the official implementation's additional batch divisor. This matches the intended gradient scale, not the stochastic optimization trajectory or total view-sampling budget. The shortened budget was introduced after small feasibility trials produced usable final training artifacts. There is no controlled, sufficiently broad 300-versus-3,600 convergence or quality-equivalence study. More iterations could improve donor transfer or instead exacerbate artifacts. Admitted early 3,600/600-iteration variants remain in the snapshot with their original provenance; the snapshot is not a homogeneous 300-iteration benchmark.

Optional refinement and admission. Balanced JN-L separately optimizes a coarse deformation field for 120 Adam iterations at learning rate 0.003. It combines Jacobian and source-crease-aware normal terms with an area-weighted residual umbrella Laplacian. For a frozen coarse guide $\bar{V}$ and residual $r_i = v_i - \bar{v}_i$, this term is

$$E_L = \sum_i m_i \left\| r_i - \sum_{j \in \mathcal{N}(i)} w_{ij} r_j \right\|_2^2 . \tag{9}$$

Source-graph inverse-edge-length weights are row-normalized, with boundary, nonmanifold and sharp-crease edges excluded. The latter term's weight is fixed from the initial gradient-norm ratio to the weighted J+N terms, without outcome-based weight search. Weak proposal-fidelity/coarse terms and source assembly constraints stabilize the solve; proposal fidelity is not a final admission requirement. PA and its refined output are both retained if independently usable and geometrically distinct. The final 10,165-example snapshot contains 5,611 PA, 4,548 balanced JN-L and six legacy/bootstrap examples. These counts include different inputs and configurations and are not a causal comparison of the two arms.

Cheap geometry and deformation diagnostics precede expensive rendering and encoding. Diagnostic cards use a small fixed RGB subset (typically three views) and eight clay views from the common 24-camera convention; a full 24-view Cycles sweep is not required for every candidate. Mesh-level naturalness and view-level edit visibility are assessed separately. The publication audit read and collated all 10,165 Shape512 and all 10,165 SS examples. Higher-resolution Shape/PBR supervision is prepared separately for the native cascade used in this paper. Loader success establishes usable artifacts, not a calibrated naturalness probability. Canonical face correspondence belongs to the stored mesh pair; it is not asserted for surfaces decoded by the native VAEs.

All snapshot entries record *human_review=false*; screening was model-assisted. Selection-biased quality spot checks do not estimate the population error rate. Residual small intersections, attachment gaps and local folds may remain, and ISIR success is not a prerequisite for inclusion. The construction is intended

to supply plausible, visible shape changes, not certified CAD-quality assemblies. Its source categories are long-tailed (2–592 edits per category), rather than category-balanced.

Rendering, provenance and split boundary. The canonical M0 frame and source UV/materials are shared by M0 and each final M1. Mesh/attribute, source/donor, configuration and worker hashes, diagnostics, render evidence and loader receipts accompany the sample identity. The donor is absent from retargeting inputs: an edited M1 view provides the desired shape and undeformed M0 views supply appearance. Exact face/barycentric surface correspondence can supervise matching; it does not supply ground-truth surface visibility at inference.

The logical split uses seed 20261012 and groups all variants of every source UID and existing geometry cluster. Orientation-uncertain train-only cases remain in training. UID and cluster intersections across splits are zero, but donor/source roles may cross split boundaries. The split is assigned after the approved snapshot is constructed. It does not establish donor disjointness, complete semantic-duplicate removal, independence from earlier multi-view pretraining, or an outcome-blind evaluation of GeoStyle's population-level feasibility. The fixed indices are mapped by sample identity; the original physical storage layout is not the logical split.

A.2 Architecture, matcher and training details

D1 has one query-only cross-attention block with four heads of width 32 and a 128-dimensional feed-forward bottleneck. Candidate descriptors are unchanged. A LayerNorm–MLP scalar head of hidden width 64 adds a bounded 0.25 tanh residual to contextual cosine. The list-wise objective averages eligible queries within an asset; assets without an eligible query do not update the matcher. The final seven-pass checkpoint contains 62,545 updates, using AdamW rate 10^{-4}, weight decay 0.01 and gradient clipping 1. Six teacher-rejected cases affect matcher training only; the SS population is still 9,149. No test matching-head training or calibration is performed.

The selected SS uses identity values and no new learned readout. An evaluated alternative anchors learned logit refinement to its initial score:

$$a_{rhk} = \mathrm{softmax}_{k\in\mathcal{K}_r}[\log p_r(k) + s_{rhk}(\theta) - s_{rhk}(\theta_0)]. \quad (10)$$

Four-head probabilities average before native value aggregation. Prior-weighted centering of the score difference preserves this softmax, and zero prior support remains zero. This initialization reproduces Fixed weights, but an initialization safeguard is not evidence of a quality advantage. Learned V/output transforms are distinct alternatives. Absent deformation-query channels and source modules are physically removed in their respective controls.

The original SS/PBR base and frozen matcher tensors, LoRA initialization, per-rank input/noise/timestep traces, and saved checkpoint/config hashes are verified in the experimental artifacts. Main SS comparisons use a native microbatch 96 without accumulation; native PBR uses four ranks with microbatch 6, accumulation 1 and global batch 24. Shape stages, image encoders, Omega and the matching decoder remain frozen during generative fine-tuning.

A.3 Metric implementation details

Let A, B be the two sets of 20,000 area-weighted surface samples and $d(a, B) = \min_{b\in B}\|a - b\|_2$. The reported unsquared distance is

$$\mathrm{CD} = \tfrac{1}{2}\left[|A|^{-1}\sum_{a\in A} d(a,B) + |B|^{-1}\sum_{b\in B} d(b,A)\right]. \quad (11)$$

Precision is the fraction of A with distance at most 0.02 to B; recall reverses the roles. Surface F-Score is their harmonic mean, or zero when both fractions vanish. The source-defined normalization and stored GT point sample are common to models; there is no independent output alignment.

Photometric scores use the archived 512-pixel float RGB and nominal unit range. Native PSNR is $-10\log_{10}$ MSE over the full image; VGG LPIPS receives the affine normalization from $[0, 1]$ to $[-1, 1]$. SSIM uses an 11-pixel Gaussian window, standard deviation 1.5 and the native zero padding. Mask IoU thresholds both alpha maps at 0.5.

DreamSim uses the official trained ensemble; complementary CLIP uses a separate original OpenAI ViT-B/16 image encoder. Their paths clamp RGB to $[0, 1]$, round to nearest 8-bit RGB, and apply their respective official preprocessing. Inception receives the same full-frame 8-bit conversion. Neural metric evaluation uses FP32, TF32 off and no autocast. Source scores average cameras 9/10 within an example; camera 11 conditions generation but is not added to the scored pool. Target scores use camera 8.

Supplementary foreground PSNR uses the union of predicted and GT alpha masks above 0.5. It averages RGB squared error only on that union and clamps the resulting MSE below at 10^{-12} before taking the logarithm. This is not a common GT-only mask. The separate GT crop is the GT-alpha bounding box with padding $\lceil 0.1\max(w, h)\rceil$, clipped to the image boundary; the identical box crops prediction and reference for every model. It can retain background inside the rectangle and does not align or recenter individual predictions.

The edited GT mesh is rasterized independently. After a post-TEST GT-only audit of all 1,524 views, the one-pixel-dilated raster gate permits at most $\lceil 0.01 N_{\mathrm{GT}\alpha}\rceil$ outside pixels. The original continuous 0.99 containment gate rejected one thin-object view (7/648 outside pixels; containment 0.9891975); the uniform integer rule admits

all views. The failed run is retained. Only this admission gate was adapted in rendering and case validation; GT, predictions, metric formulas and frozen model selection are unchanged. The gate is not a score.

A.4 Original module family and independent training

The original 12-model family has ten prespecified module contrasts; its adjusted intervals include zero. Two follow-up evidence designs are reported separately. In the original FULL family, ray minus voxel macro IoU is -0.000236, with adjusted interval $[-0.000878, 0.000351]$. Member minus mean is -0.000601, with $[-0.002246, 0.001262]$. These results do not establish a positive quality effect or equivalence, and the original learned readout is not the selected Fixed architecture.

The two-seed SS study uses 4,000 source-cluster draws for three IoU contrasts separately within each seed and for the fixed-two-checkpoint mean. It is not a simultaneous nine-comparison or training-seed-population interval. Fixed minus V1 is positive in both actual runs; Fixed minus Top-1 remains uncertain. For selected Fixed SS, two fresh VAL36 loads verify latent/occupancy consistency and tail-batch behavior before TEST. The V1/Top-1 controls retain their own archived checkpoint-binding and native parity checks; these are not a two-fresh-load study of all seven ablation checkpoints.

The two-seed PBR training diagnostic retains each producer's fixed-four GT-shape target reference and declared initial-render policy. Across seeds, input RGB, cameras, noise and initial latent checks are exact, but these curves are not pooled as common-pixel source-preservation evidence.

A.5 Paired geometric and appearance studies

All 48 downstream family values are retained unrounded in the source-bound artifact. The displayed geometric intervals use that complete correction; the separate SS diagnostic family averages three noises per asset. The historical V1-geometry PBR table and the same-selected-Fixed table in the Main text describe different fixed geometries. Their points are not exchanged or pooled. Generated-geometry appearance and geometry results remain distinct from fixed-GT-shape training diagnostics.

A.6 Matching evidence and its limits

On 31,408 native observable Top-1-miss/Top-K-hit relations, mean GT mass is 1.6266% for post-hoc Uniform, 6.4601% for Fixed, and 6.1756% for learned refinement. The learned-minus-Fixed interval includes zero. Candidate diagnostics and a relationship's stored annotation do not establish a population quality gain or a visibility calibration. The illustration is selected by the original conditional lexicographic rule, not downstream render quality. Post-hoc Uniform mass is not the matched Uniform training arm.

A.7 Full-frame and foreground sensitivity

The primary table retains full-frame metrics. The existing producer's supplementary scores on identical selected Fixed geometry show that Top-1 R minus vanilla changes source DreamSim by -0.000517 full-frame and -0.000893 on the common GT crop. Source CLIP instead changes by -0.000487 and -0.000270, respectively. Foreground PSNR changes by -0.01205 dB at source cameras and $+0.06557$ dB at the target. Thus the small primary appearance points are not a consistent advantage across perceptual similarity, semantic cosine and pixel alignment. Feature has the lowest crop DreamSim point and Raw has the highest crop CLIP point; these descriptive changes in ranking do not reselect the frozen PBR model. Absolute foreground PSNR is much lower than its full-frame value because the pixel support differs; a high full-frame PSNR alone does not establish foreground fidelity. These are existing supplementary endpoints, not a new primary family or additional independent confirmation.

A.8 Qualitative selection and remaining examples

The original metadata declaration fixes two examples from distinct sources without candidate scores. The first is placed in the Main text; the second and every conditional lower-median example are included here. Empty categories remain empty and overlapping categories share one panel. Category counts describe eligibility, not population success rates. Images reuse exactly the scored float arrays and official uint8 conversion, with target8/source9/10, common M1 original-material references, and no model-specific crop or rerender.

Predeclared conditional eligibility. joint-improvement: 77 eligible. The selected lower-median example appears in Fig. 9. geometry-loss: 191 eligible. The selected lower-median example appears in Fig. 10. appearance-loss: 153 eligible. The selected lower-median example appears in Fig. 11. Overlapping conditions share one panel. The first outcome-blind panel stays in the Main text even when it also satisfies a conditional category; it is referenced here rather than duplicated.

A.9 Execution and reproducibility scope

All 5 fixed pipelines use 508 test examples from 134 source clusters and one prescribed generation noise per example. Global occupancy ratios are distinguished from mean per-asset IoU. All 2,540 SS archives are checked for actual

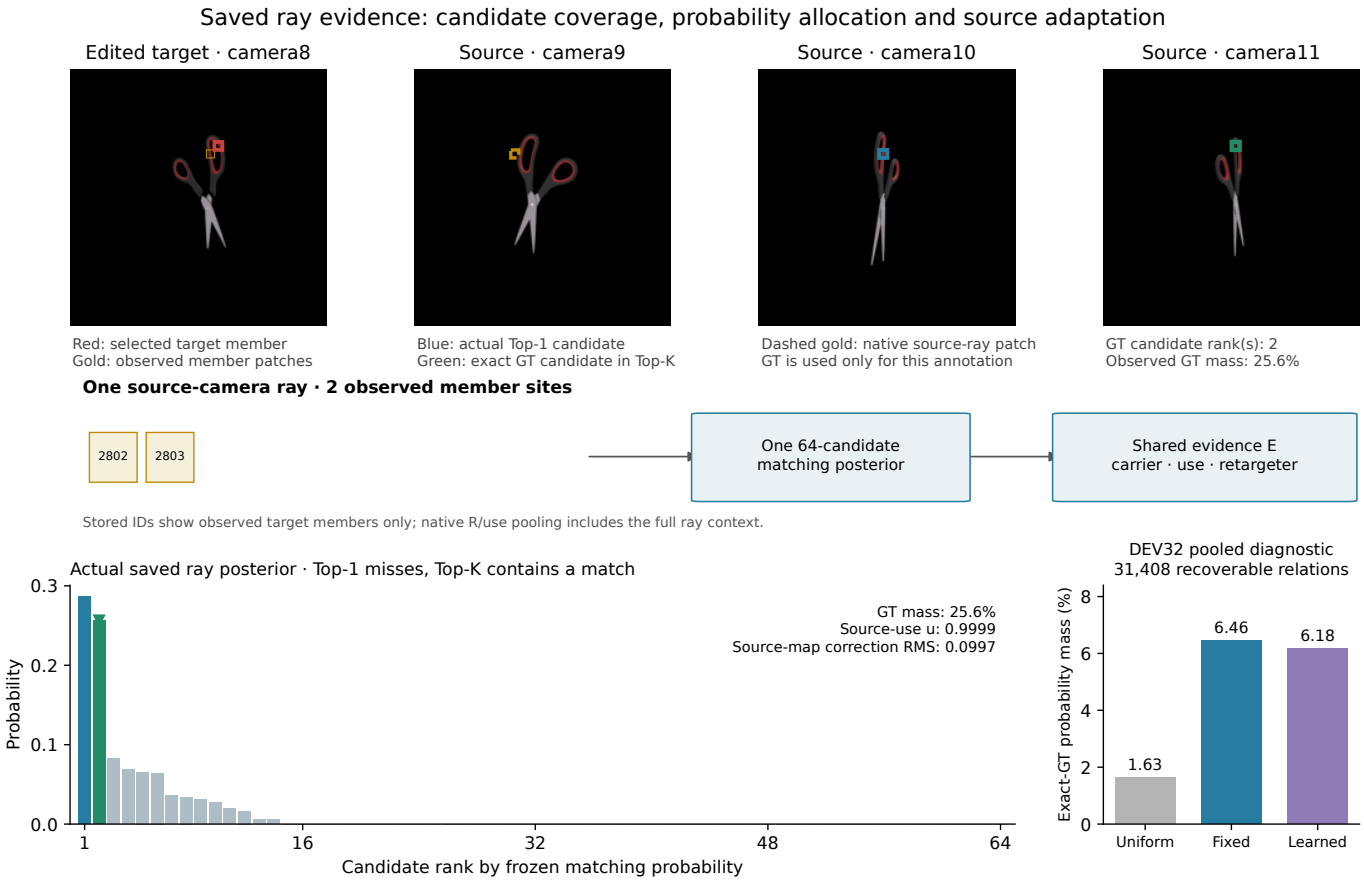


Figure 7: Actual saved native ray evidence from the development panel. The top row shows the edited target and three original source inputs under the native black-background preprocessing. Red and gold boxes mark the selected target patch and observed target members; the dashed gold box marks the source-camera ray patch. Blue marks the actual Top-1 proposal and green the exact GT candidate retained at rank two. The two stored member sites are observed target members, not the entire ray context pooled by the retargeter and source-use branch. The fixed matching posterior assigns 25.64% mass to the GT candidate in this example. The bottom-right diagnostic averages exact GT mass over 31,408 native Top-1-miss/Top-K-hit relations from 32 assets; Uniform is post-hoc reweighting, not the matched Uniform training control. The illustrated relation is the first eligible relation in lexicographic sample order, requiring at least two observed sites, without selection by downstream quality. GT is used only for annotation and diagnostic selection. These conditional observations do not establish generated-quality gains, success frequency, or calibrated visibility.

SS model	Macro voxel IoU↑
V1	0.467355
C_SPATIAL	0.471242
C_RAW	0.472360
C_D1_COS	0.473008
C_DECODER	0.472600
C_DECODER_NO_R	0.468671
C_TOP1_V	0.472218
C_FIXED_TOPK	0.472745
FULL	0.472078
FULL_QK_ONLY	0.472446
FULL_VOXEL_Q	0.472314
E_FIXED_RAW	0.473471
E_LEARNED_RAW	0.474794
FULL_MEAN_MATCH	0.472679

Table 5: Fourteen completed SS endpoints on DEV128, 93 source clusters, three generation noises, training seed 101, and 480 updates. The noises are averaged per example. The original twelve-model comparison family is complete; its ten module contrasts are reported separately. Mean/member is a conditional comparison in the original FULL family. The learned-evidence endpoint belongs to the separately reported exploratory follow-up. FULL is an implementation identifier, not the final selected model.

Original study contrast	Paired IoU change	Ten-contrast family CI95
Spatial to raw semantic	+0.001117	[-0.003255, +0.004969]
Raw to decoder descriptor	+0.000648	[-0.003386, +0.005441]
Decoder cosine to scalar score	-0.000408	[-0.002740, +0.001883]
Value transform	-0.000382	[-0.004135, +0.003479]
Top-1 to fixed Top-K	+0.000527	[-0.000955, +0.001938]
Learned attention	-0.000667	[-0.002299, +0.000777]
Original matching-prior control	-0.000368	[-0.002372, +0.001694]
Ray versus voxel query	-0.000236	[-0.000878, +0.000351]
Retargeting at Top-1	+0.003929	[-0.000525, +0.008987]
Member versus mean proposal	-0.000601	[-0.002246, +0.001262]

Table 6: Completed original twelve-model DEV128 experiment, training seed101, three paired generation noises. All ten prespecified module-family adjusted intervals include zero. Contrasts retain their original contexts: retargeting is tested at Top-1, ray queries in the original learned readout, and the original prior control also changes utility inputs. Spatial-to-semantic substitution changes support from spatial3 to semantic64. These results do not replace final-context matched controls of the shared identity-value design.

Training seed	Vanilla IoU	Top-1 IoU	Fixed IoU	Fixed minus V1 [CI95]	Fixed minus Top-1 [CI95]
101	0.467355	0.472600	0.473471	+0.006116 [0.001139, 0.011309]	+0.000871 [-0.001356, 0.003387]
202	0.467177	0.472346	0.474730	+0.007553 [0.002225, 0.013285]	+0.002385 [-0.000206, 0.005183]
Mean of two	0.467266	0.472473	0.474101	+0.006835 [0.001834, 0.012196]	+0.001628 [-0.000477, 0.003957]

Table 7: Completed SS confirmation on the same DEV128 assets and 93 source clusters, averaging three paired generation noises per asset. Each seed and the fixed-two-seed mean use the declared three-contrast IoU family separately: Top-1 minus V1, Fixed minus V1, and Fixed minus Top-1. Intervals use 4,000 paired source-cluster draws and condition on the trained checkpoints; they do not estimate a population of training seeds or form a simultaneous nine-comparison interval. These are development SS results, not retargeter-only effects, model selection, or held-out cascade quality.

latent hashes, packed occupancy and TP/FP/FN consistency; occupancy is not substituted for surface CD/F-Score or appearance metrics. Generated geometry is not aligned independently per output. PBR comparisons retaining the same geometry use identical original guide values/order/scale and rendering triangles.

The test campaign bounds complete-asset disk credits. Local raw and fully consumed guide duplicates are offloaded only after producer/reader checks and fresh canonical file-server hashes. Scored pixels and provenance remain local, and canonical intermediate archives remain on the file server. Completed native cases

Seed	Model	Target PSNR↑	Crop LPIPS↓	Crop SSIM↑	Base-color PSNR↑
101	Vanilla	26.876521	0.086454	0.953675	25.254020
101	Top-1 R	26.860112	0.085186	0.954216	25.402984
202	Vanilla	26.862564	0.086446	0.953372	25.168122
202	Top-1 R	26.618615	0.082854	0.953820	25.285976

Table 8: Completed PBR training diagnostic at 1,900 updates for two training seeds. Each row uses four target-camera examples with fixed GT shape; PSNR uses pooled pixel MSE. Within each seed, V1/R and all six validation epochs share exact GT pixels. Independent seeds retain their own rendered references, so no cross-seed quality pooling is performed. This small training-QA panel does not establish inferential rendering quality, source preservation, generated geometry, or pipeline generalization.

Contrast	CD change	Adjusted interval	F-Score change	Adjusted interval
Raw semantic minus spatial	-0.916	[-4.133, +2.882]	+1.698	[-4.555, +9.167]
Decoder minus raw semantic	-0.724	[-3.254, +1.600]	+1.383	[-5.267, +7.915]
Scalar minus decoder cosine	-0.493	[-1.610, +0.620]	+1.644	[-1.731, +5.399]
Fixed Top-K minus Top-1	-0.223	[-2.275, +2.597]	+3.963	[-0.910, +9.748]
Fixed minus uniform	-0.732	[-2.695, +1.052]	+1.553	[-3.326, +7.439]
Fixed minus physical no-R	-1.990	[-6.316, +4.024]	+9.020	[-1.582, +20.293]

Table 9: Completed matched geometry contrasts on DEV128 (93 source clusters), one training seed and one paired generation noise. Intervals retain the original48-comparison family across six contrasts and eight endpoints, using48,000 paired source-cluster draws; only the two geometry endpoints are displayed. CD differences and intervals are in units of0.0001 and F-Score in0.001. All intervals include zero. Positive F-Score and negative CD points are descriptive; they do not establish module necessity or equivalence. This development study is distinct from the frozen final-test comparison.

Completed Fixed-context contrast	IoU change	Unadjusted CI95	Six-contrast CI95
Raw semantic minus spatial	+0.000708	[-0.002025,0.003354]	[-0.002955,0.004177]
Decoder cosine minus raw	+0.000476	[-0.001446,0.002465]	[-0.002116,0.003196]
Scalar Fixed minus decoder cosine	+0.000021	[-0.000645,0.000712]	[-0.000872,0.000973]
Fixed minus actual Top-1	+0.000871	[-0.001025,0.002918]	[-0.001642,0.003679]
Fixed minus Uniform-K	-0.000142	[-0.001965,0.001611]	[-0.002590,0.002233]
Fixed minus physical no-R	+0.004639	[0.000565,0.009024]	[-0.000811,0.010633]

Table 10: Completed seven-model SS study and all six prespecified development contrasts. All use actual 480-update checkpoints, the same 128 assets from 93 source clusters, and three paired generation noises averaged within each asset. Percentile intervals use 12,000 paired source-cluster resamples and the original six-comparison Bonferroni correction. All adjusted intervals include zero. Positive point estimates do not establish component necessity; these SS results also do not establish equivalence or full-cascade benefit.

Actual DEV32 composition	Source LPIPS↓	Source DreamSim↓	Source CLIP↑	Target PSNR↑	Target SSIM↑	Target M-IoU↑	CD↓	F-Score↑
Vanilla V1	0.053099	0.155692	0.942434	24.47068	0.950488	0.880945	0.012452	0.849149
Top-1 retargeting	0.052104	0.152435	0.940636	24.57466	0.951284	0.885678	0.012413	0.851420
D1-cos SS + Top-1 R PBR	0.053362	0.155373	0.942157	24.57729	0.951170	0.886600	0.012504	0.851308
Fixed-evidence SS + Top-1 R PBR	0.053290	0.157828	0.941692	24.54433	0.951563	0.886574	0.012328	0.856566
Anchored-learned SS + Top-1 R PBR	0.052894	0.156364	0.943132	24.62475	0.951161	0.885769	0.012498	0.854233

Table 11: Actual DEV32 compositions, one training seed and one generation-noise realization. The source reference is original material on M1 geometry. The same V1 and retargeting per-asset baselines are verified across the three candidate studies. This is a development table; FID/KID and held-out test results are not included.

from a stopped campaign retain their original paths, model/noise bindings, terminal records and scored-image identities; remaining cases use the same frozen numerical protocol. Protected readers cause waits, and retirement transactions do not signal GPU workers. Recovery changes scheduling and cache lifecycle, without choosing models from test scores or altering metric endpoints. The final metric and qualitative artifacts retain actual producer/file hashes and the original model-freeze/declaration lineage.

	Source preservation				Target alignment			
SS variant	LPIPS↓	DreamSim↓	CLIP↑	PSNR↑	SSIM↑	M-IoU↑	CD↓	F-Score↑
Top-1 R	0.053227	0.148809	0.939061	24.07381	0.948434	0.890535	0.0127795	0.840095
Fixed	0.053438	0.151909	0.940282	24.01288	0.948494	0.892593	0.0127572	0.844058
Learned	0.053275	0.151001	0.940279	24.04810	0.948579	0.890776	0.0128053	0.842134

Table 12: Actual paired DEV128 (93 source clusters), one training seed and one generation-noise realization. All pipelines use the same frozen Shape stages and trained Top-1 R PBR. Source cameras9/10 use the original-material-on-M1 reference; target alignment images use camera8. Geometry uses the generated surface without output alignment. The original32 cases are reused and96 added per model; this is an exploratory extension, not held-out confirmation.

PBR arm on identical V1 geometry	LPIPS↓	DreamSim↓	CLIP↑	Target PSNR↑	Target SSIM↑
Vanilla	0.053099	0.155692	0.942434	24.47068	0.950488
Actual Top-1 + R	0.052984	0.154160	0.942141	24.54852	0.950585
Top-1 + learned feature + R	0.053019	0.154338	0.942222	24.51148	0.950536
Raw Top-K + learned feature + R	0.052986	0.154346	0.942330	24.52061	0.950544

Table 13: Actual appearance-stage ablation on32 development examples. Only PBR changes: generated V1 geometry, native guides, initial texture noise, sampler, actual rendering triangles, and material references are paired. LPIPS/DreamSim assess source appearance; CLIP is complementary and target image metrics are separate. CD/F-Score are identical by construction. The feature/Raw arms keep a Top-1 carrier and source-use path; these are not matched shared-evidence Fixed PBR controls. Final PBR selection requires the chosen SS geometry, rather than transferring this V1-geometry ranking.

PBR on the same E-fixed SS	Source LPIPS↓	DreamSim↓	CLIP↑	Target PSNR↑	SSIM↑	M-IoU↑
Vanilla	0.053513	0.158346	0.942179	24.47775	0.951534	0.886571
Top-1 R	0.053290	0.157828	0.941692	24.54433	0.951563	0.886574

Table 14: PBR-only DEV32 comparison on identical fixed-evidence SS geometry, native guides, initial texture noise, render triangles, and M1 references. CD=0.0123276 and F-Score=0.856566 are identical by construction. The source LPIPS and target PSNR intervals include zero. This is not a comparison between different SS models.

Matching diagnostic	Exact@1	Exact@8	Exact@64	Near-1@64
Raw cosine, mean-ray support	1.25	11.09	50.76	80.41
D1 cosine, same support	7.73	28.47	50.76	80.41
D1 scalar score, same support	8.56	29.92	50.76	80.41
Member-posterior mixture	11.19	45.68	73.36	95.59

Table 15: Reused matching diagnostics on 50,493 observable source-camera-ray/target-patch relations from 32 development source clusters (percent). Exact denotes the GT source patch on the 32-by-32 image grid; Near-1 allows a one-patch neighborhood. These are conditional relation-coverage measures, not per-point pixel accuracy or generative-quality scores. GT visibility is used only to define the diagnostic denominator, not as an inference write gate.

A. Source appearance: full frame and common GT-defined crop				
PBR arm	DreamSim full↓	DreamSim GT crop↓	CLIP full↑	CLIP GT crop↑
Vanilla	0.158346	0.194194	0.942179	0.935360
Actual Top-1 + R	0.157828	0.193302	0.941692	0.935090
Top-1 + learned feature + R	0.157985	0.193285	0.941759	0.935348
Raw Top-K + learned feature + R	0.158261	0.193599	0.941559	0.935625
B. Pixel alignment: full frame and alpha-union foreground				
PBR arm	Source PSNR full↑	Source PSNR foreground↑	Target PSNR full↑	Target PSNR foreground↑
Vanilla	23.68253	11.62652	24.47775	12.34999
Actual Top-1 + R	23.67047	11.61447	24.54433	12.41556
Top-1 + learned feature + R	23.65548	11.59947	24.51012	12.38134
Raw Top-K + learned feature + R	23.66457	11.60856	24.52268	12.39391

Table 16: Existing supplementary scores on the same selected Fixed geometry, 32 development examples and 32 source clusters. Source cameras9/10 are averaged per example; target uses8. The GT-alpha bounding box, padded by10% of its larger side, defines one common prediction-independent crop; it may still contain background. Foreground PSNR instead averages RGB squared error over the union of predicted/GT alpha> 0.5, so its mask can vary by output. Full-frame PSNR/SSIM and the primary source LPIPS/DreamSim retain their original definitions. These descriptive sensitivities neither change the primary endpoint/family nor reselect the frozen PBR checkpoint.

A. Source appearance: full frame and common GT-defined crop				
Model	DreamSim full↓	DreamSim crop↓	CLIP full↑	CLIP crop↑
Zero-shot, source-only RGB4	0.153975	0.184780	0.943278	0.934490
Zero-shot, edited target + source RGB3	0.154276	0.186046	0.942090	0.935730
Vanilla V1	0.151460	0.184041	0.942854	0.937057
Top-1 R	0.151162	0.183377	0.943177	0.936402
Selected model	0.152069	0.183833	0.943018	0.935886
B. Pixel alignment: full frame and alpha-union foreground				
Model	Source PSNR full↑	Source PSNR foreground↑	Target PSNR full↑	Target PSNR foreground↑
Zero-shot, source-only RGB4	23.17341	11.74087	22.40621	10.94126
Zero-shot, edited target + source RGB3	23.46619	11.90479	23.54157	11.78260
Vanilla V1	23.62321	12.05047	23.83599	12.06429
Top-1 R	23.70646	12.11024	23.99656	12.18422
Selected model	23.71804	12.11966	24.02827	12.21706

Table 17: Existing supplementary scores for the complete frozen TEST508 pipelines. Source means use cameras9/10; target means use camera8. Crops use the common GT-alpha bounding box with10% padding. Foreground PSNR uses the union of predicted/GT alpha> 0.5 and can have output-specific support. These descriptive composed-pipeline sensitivities do not isolate a PBR effect, change the primary family, or reselect a checkpoint.

Contrast	Endpoint	Difference	Family-adjusted95% interval	Interpretation
Selected minus Vanilla V1	LPIPS	-0.001044	$[-0.001715, -0.000446]$	benefit supported
Selected minus Vanilla V1	DreamSim	0.000609	$[-0.002974, 0.003841]$	uncertain
Selected minus Vanilla V1	CLIP	0.000164	$[-0.002010, 0.002200]$	uncertain
Selected minus Vanilla V1	PSNR	0.1923	$[0.0703, 0.3201]$	benefit supported
Selected minus Vanilla V1	SSIM	0.001156	$[0.000538, 0.001833]$	benefit supported
Selected minus Vanilla V1	M-IoU	0.012041	$[0.005487, 0.021451]$	benefit supported
Selected minus Vanilla V1	CD	-0.0005139	$[-0.0014386, 0.0000208]$	uncertain
Selected minus Vanilla V1	F-Score	0.006059	$[-0.000676, 0.013572]$	uncertain
Selected minus Top-1 R	LPIPS	-0.000129	$[-0.000530, 0.000291]$	uncertain
Selected minus Top-1 R	DreamSim	0.000907	$[-0.001656, 0.003761]$	uncertain
Selected minus Top-1 R	CLIP	-0.000159	$[-0.002212, 0.001829]$	uncertain
Selected minus Top-1 R	PSNR	0.0317	$[-0.0653, 0.1267]$	uncertain
Selected minus Top-1 R	SSIM	0.000237	$[-0.000321, 0.000673]$	uncertain
Selected minus Top-1 R	M-IoU	0.002863	$[-0.000239, 0.008191]$	uncertain
Selected minus Top-1 R	CD	-0.0000233	$[-0.0001668, 0.0000903]$	uncertain
Selected minus Top-1 R	F-Score	-0.000226	$[-0.003362, 0.002879]$	uncertain

Table 18: All predeclared selected-versus-control primary comparisons, using the existing source-cluster-adjusted intervals. Source LPIPS/DreamSim/CLIP use9/10; target PSNR/SSIM/M-IoU use8; CD/F-Score use generated surfaces. Values are rounded for display; interpretation uses the original unrounded intervals and metric directions. An interval containing zero does not prove equivalence. FID/KID remain descriptive and are not included in this inferential family.

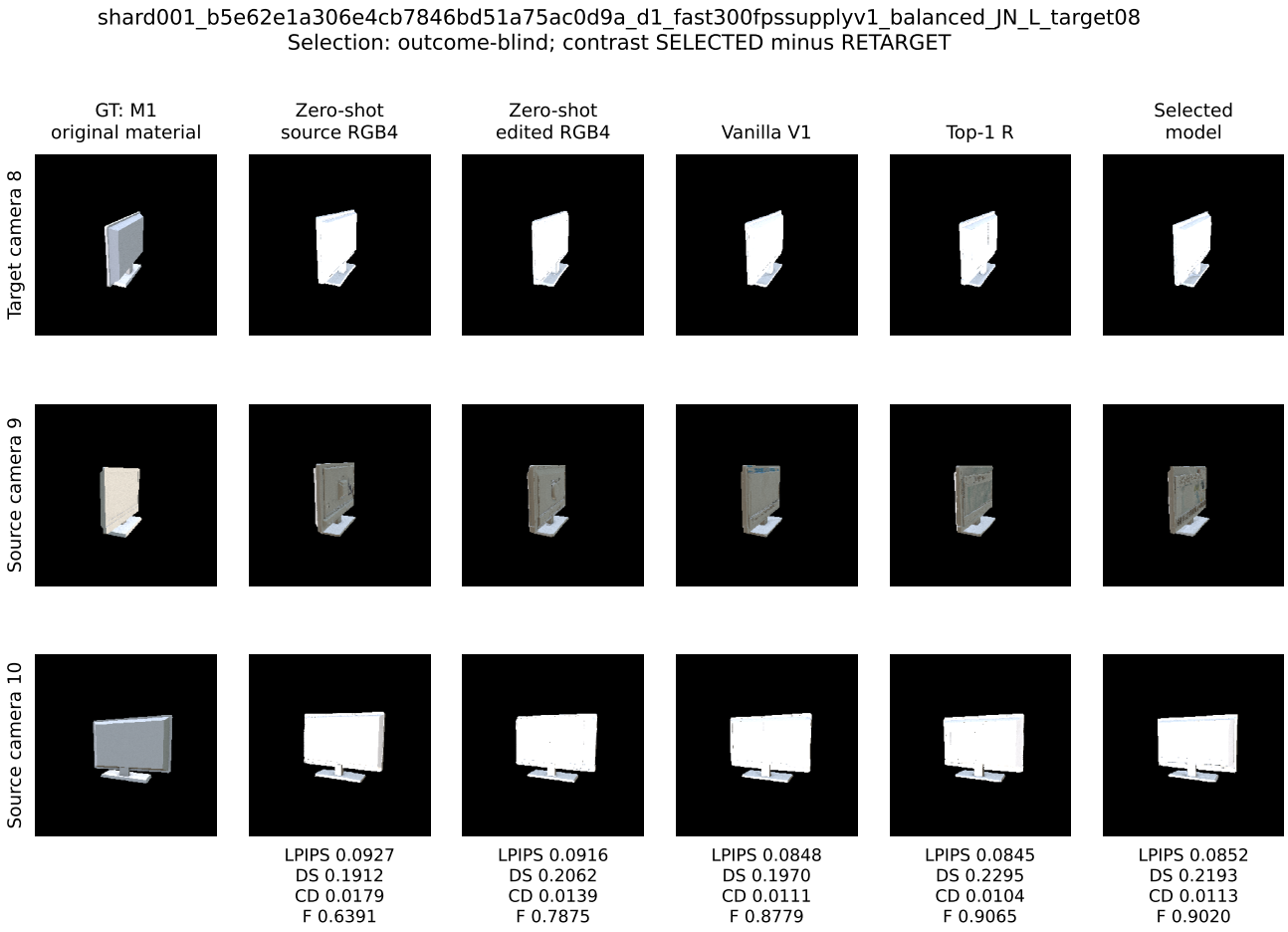


Figure 8: Actual scored frozen TEST example. Rows are target8/source9/source10; columns are the common original-material M1 reference and every frozen model. Selection: outcome-blind. Full-frame panels use the scored archive and official uint8 conversion, without model-specific crop, alignment or rerender. Source annotations use cameras9/10. Conditional categories illustrate their stated conditions and do not estimate population success rates.

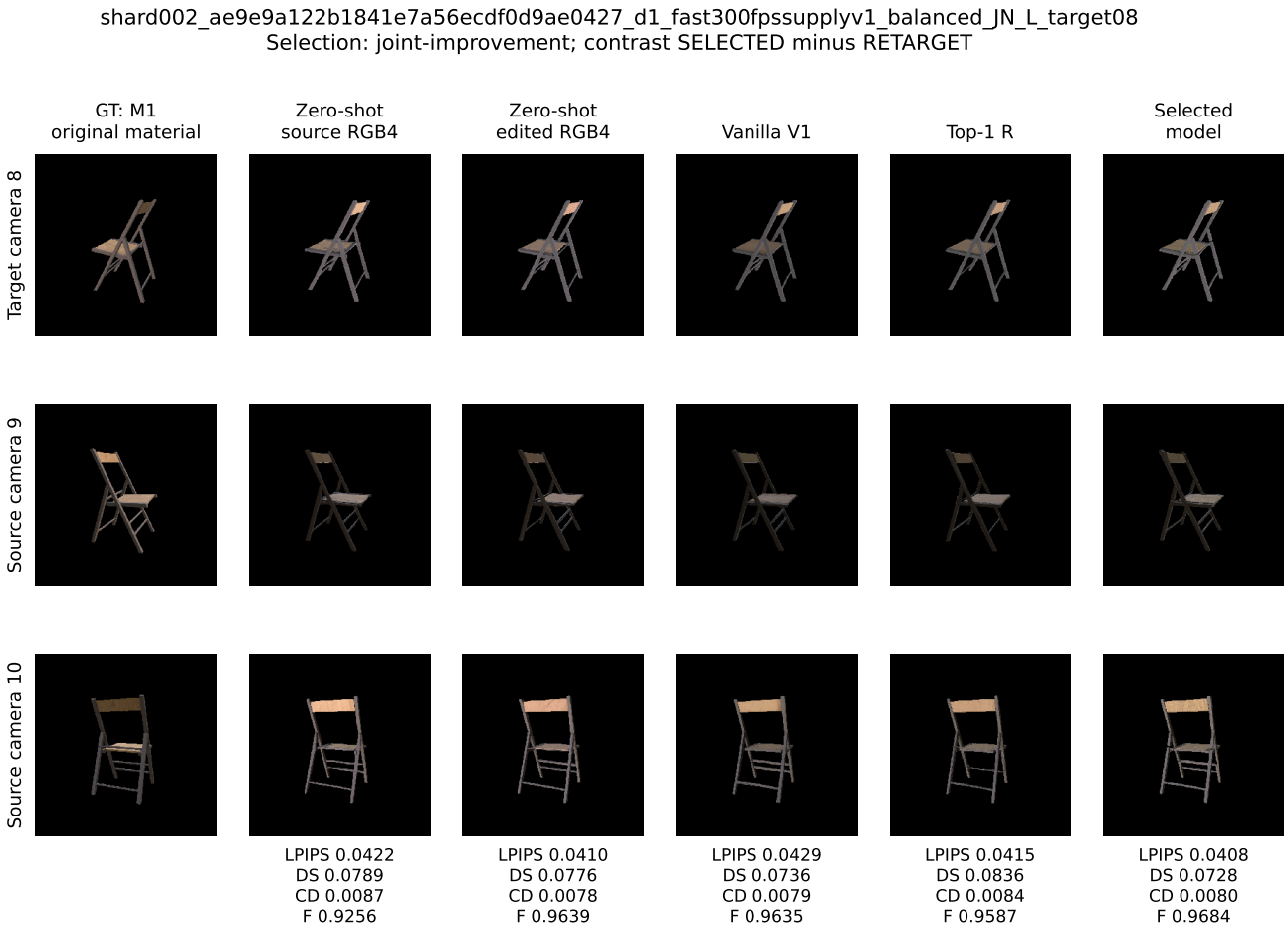


Figure 9: Actual scored frozen TEST example. Rows are target8/source9/source10; columns are the common original-material M1 reference and every frozen model. Selection: joint-improvement. Full-frame panels use the scored archive and official uint8 conversion, without model-specific crop, alignment or rerender. Source annotations use cameras9/10. Conditional categories illustrate their stated conditions and do not estimate population success rates.

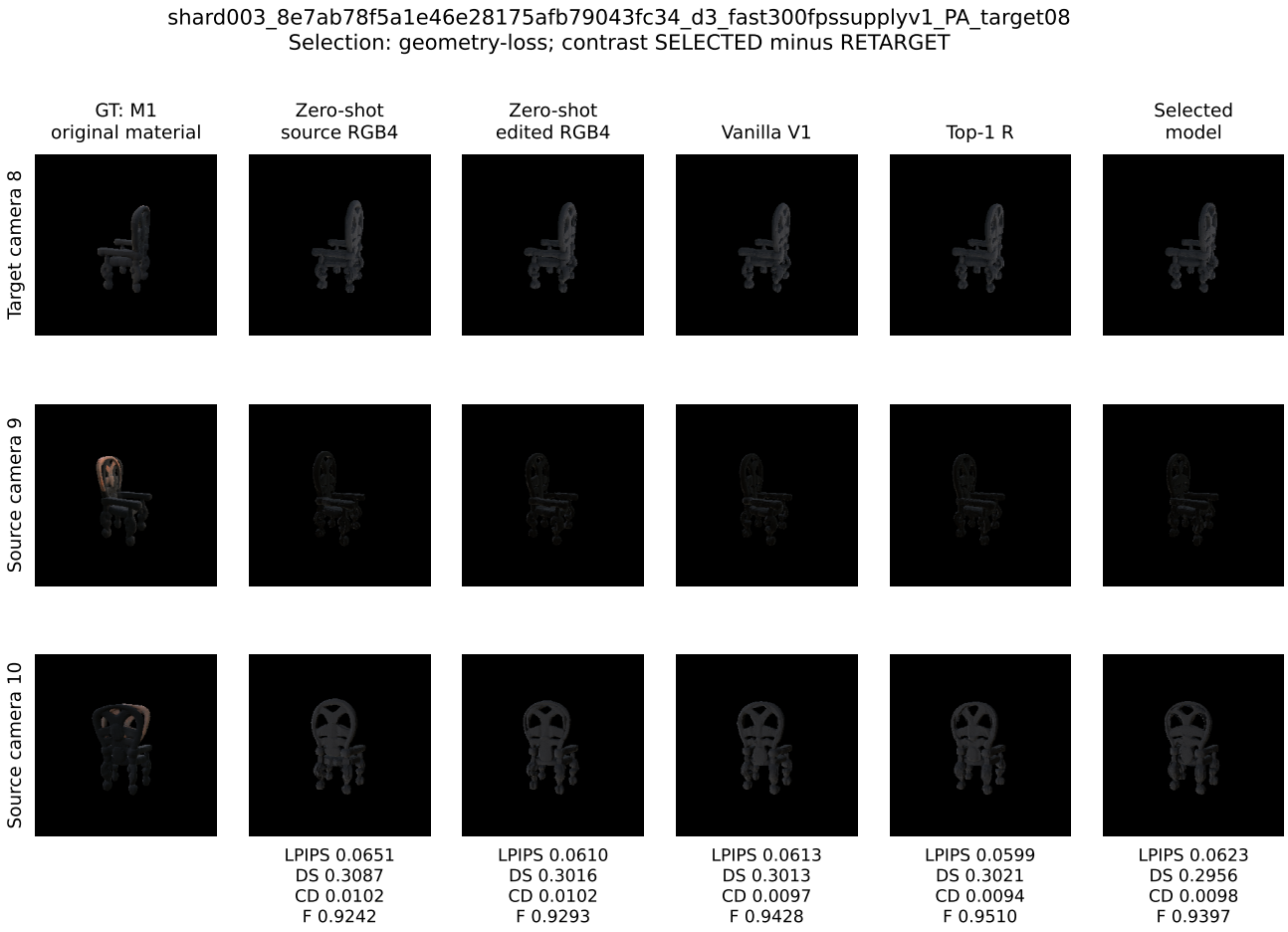


Figure 10: Actual scored frozen TEST example. Rows are target8/source9/source10; columns are the common original-material M1 reference and every frozen model. Selection: geometry-loss. Full-frame panels use the scored archive and official uint8 conversion, without model-specific crop, alignment or rerender. Source annotations use cameras9/10. Conditional categories illustrate their stated conditions and do not estimate population success rates.

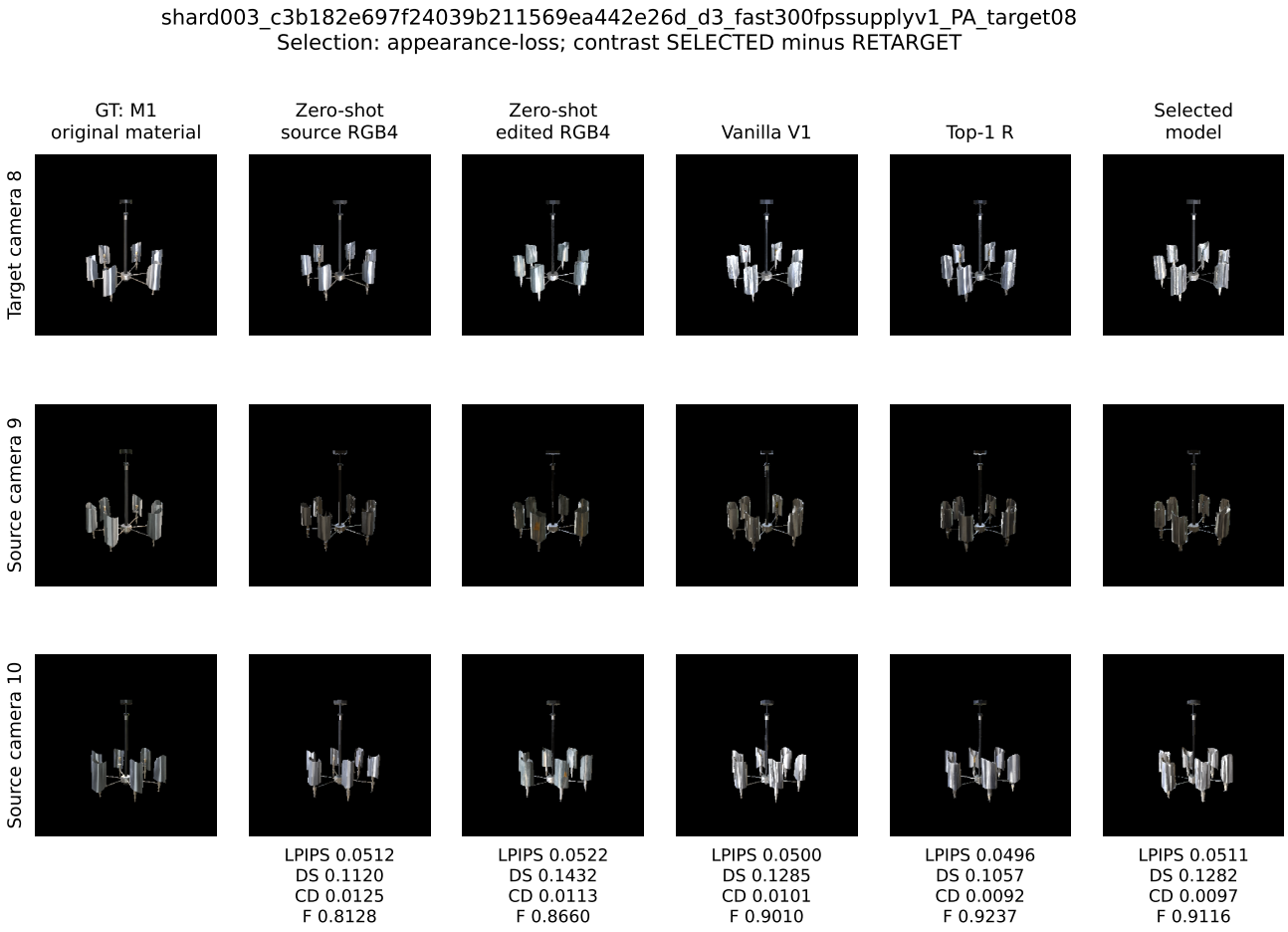


Figure 11: Actual scored frozen TEST example. Rows are target8/source9/source10; columns are the common original-material M1 reference and every frozen model. Selection: appearance-loss. Full-frame panels use the scored archive and official uint8 conversion, without model-specific crop, alignment or rerender. Source annotations use cameras9/10. Conditional categories illustrate their stated conditions and do not estimate population success rates.